

SHIVAM GARG

+91-8799766316 | shivamcp2694@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Final-year Computer Science student with **9.2 CGPA** and production full-stack experience. Migrated high-traffic affiliate marketing system from Firestore to PostgreSQL at vCommission, achieving **3x performance improvement**. Expertise in Go, React, Python, and TensorFlow with published AI research and proven ability in scalable system optimization.

EDUCATION

Bennett University, Greater Noida

2022 – 2026

B.Tech - Computer Science and Engineering - CGPA: 9.2/10.0

EXPERIENCE

BVC E Services Pvt Ltd (a VC One Group company)

Aug 2025 - Present

Full Stack Developer Intern (for vCommission)

Gurugram, India (Onsite)

- Engineered complete backend migration from **Firestore to PostgreSQL** for **VelocityX**, an enterprise affiliate marketing platform managing 1000+ high-value offers across Amazon sellers and D2C brands, designing relational schema and building **REST APIs in Go (Gin)** that achieved **3x performance improvement** in system response times.
- Developed production-grade **admin dashboard with React/TypeScript**, implementing bulk product upload system processing thousands of records, integrated bug tracking with role-based access control, and advanced analytics with filtering/export capabilities.
- Optimized frontend codebase by refactoring legacy **React** components and resolving critical bugs, resulting in **40% reduction in API data fetching times** and improved user experience across affiliate partner dashboards.
- Managed production deployment pipeline using **Firebase and Google Cloud Platform (GCP)**, implementing **CI/CD automation** for continuous delivery, conducting database synchronization between environments, and ensuring zero-downtime migrations.
- Collaborated with 5-member cross-functional team following **Agile methodology**, delivering features in 2-week sprints while contributing 200+ documented hours across authentication systems, reporting endpoints, and D2C platform development.

PROJECTS

Source Sync - Narrative Analysis Tool | Python, Flask, React, TensorFlow

2025

- Built a full-stack AI analysis tool to identify and summarize narrative differences between anime adaptations and their source material using a Python/Flask backend and a React/TypeScript frontend.
- Engineered a multi-stage AI pipeline to perform semantic analysis: Leveraged **TensorFlow** for vector embeddings, **FAISS** for high-speed similarity search, and **K-Means** clustering to group unmatched content thematically.
- Leveraged a **Gemini 1.5 Flash LLM** to perform narrative analysis on thematic clusters, generating structured JSON reports on significant cut content.

AirLens - Environmental Monitoring Tool | React 19, FastAPI, PostgreSQL, Docker | 

2025

- Architected and deployed a full-stack environmental monitoring platform tracking air quality across **31+ Indian cities**, serving real-time data through a **FastAPI backend** and **React 19 frontend**.
- Developed automated **ETL pipeline** using Python and APScheduler to ingest, validate, and transform data from multiple APIs, storing **1000+ records daily** in Supabase PostgreSQL.
- Implemented intelligent caching with 1-hour TTL and server-side rate limiting, reducing API calls by **99%** and improving page load time from **800ms to 50ms**.
- Built interactive dashboard using Recharts and Leaflet maps with real-time filtering, alerts, and statistical insights across 7+ environmental metrics.
- Containerized and deployed production-ready application with **Docker**, **Nginx reverse proxy**, and **Render cloud hosting**, ensuring 99.9% uptime and CI/CD integration.

TECHNICAL SKILLS

Core: Python, JavaScript/TypeScript, Go, C++, Java, SQL, Data Structures & Algorithms

Frameworks: TensorFlow, PyTorch, React 19, Flask, Redux, Gin, Tailwind CSS, Material UI, FAISS, APScheduler, Jest, React Testing Library

Databases: MySQL, MongoDB, Supabase, PostgreSQL

Developer Tools: Git, Docker, Postman, VS Code, DBeaver

Cloud & DevOps: GCP, Firebase, Docker, CI/CD

Methodologies: Agile/Scrum, REST API Design

Certifications

- Convolutional Neural Networks in TensorFlow** – Coursera
- Google AI Essentials** – Offered by Google on Coursera

ACHIEVEMENTS and PUBLICATIONS

- Conference Publication:** Presented research paper titled *"Hybrid CNN-RNN Deepfake Detection Framework Using ResNeXt-LSTM with Temporal and Spatial Artifact Analysis"* at the International Conference on Image, Vision and Computing (ICIVC) 2025.
- Dean's List Award:** Received the Dean's List Award for outstanding academic achievement during the academic year 2024–2025.